Review changelog: batch p8 (08-testing-and-quality, lessons 01-17)

Date: 2026-10-22. Reviewer environment: Node 24.16, sandbox `/Users/todorbra/Projects/ReactLearning/tools/sandbox` (React 19.3.1, RTL 16.3.3, DTL 10.4.2, user-event 14.6.8, jest-dom 7.0.1, Vitest 3.2.7, jsdom 30.1.3, MSW 2.15.0, TanStack Query 5.104.5, Zustand 5.0.16, Zod 4.6.7). Extra installs in `scratch/p8/*`: Vitest 4.1.12 and 5.0.5 (+jsdom 30, MSW 2), react-router 7.18.4 and 8.5.0, vitest-axe 0.1.0, axe-core 4.14, @axe-core/playwright 4.14, axe-playwright 2.2.2, @playwright/test 1.64.1, happy-dom, TypeScript 7.0.2, ESLint 10.13 with typescript-eslint 8.72, react-hooks 7.1.1, jsx-a11y 6.10.2, testing-library plugin 7.16.2, @vitest/eslint-plugin 1.6.27. Scratch tests live in `/Users/todorbra/Projects/ReactLearning/tools/sandbox/scratch/p8/`.

All 170 `⚠` markers in the 17 lessons were resolved (verified, corrected, or rewritten as plain statements); none remain. A number of generic "check the docs" hedges in link tables were also removed where the URL was fetched (HTTP 200). The stale `.pdf` files in the lesson folder were not regenerated (per brief).

Totals: claims checked ~150; confirmed ~85; corrected ~45; hedged/noted ~15; unresolved 8 (listed at the end).

Most significant corrections (cross-cutting)

1. **Vitest fake timers + Testing Library hang (lessons 04, 03, 07, 16, 17, 02).** `userEvent.setup({ advanceTimers })` and `delay: null` are *not* enough under `vi.useFakeTimers()`: RTL's `asyncWrapper` (used by every `user.*`, `findBy*`, `waitFor`) ends with a real `setTimeout(0)` that is faked and only auto-advanced if a global `jest` exists (RTL `pure.js`, DTL `jestFakeTimersAreEnabled`). Reproduced: the lesson-04 SearchBox tests and the lesson-16 SearchBox test timed out at 5 s. Verified fixes: `vi.stubGlobal("jest", { advanceTimersByTime: vi.advanceTimersByTime.bind(vi) })`, `shouldAdvanceTime: true`, or faking only other timers. Examples and answers rewritten; table of variants added to lesson 04 §3.4.
2. **Vitest 3 + jsdom `AbortSignal` mismatch (lessons 06, 07, 08).** On Vitest 3.2.7 + jsdom 30, `fetch(url, { signal })` throws `TypeError: RequestInit: Expected signal ... AbortSignal`, which breaks TanStack Query (its `queryFn` signal) and React Router data routers. 9/21 of lesson-07's own example tests failed. **Not reproduced on Vitest 4.1.12 or 5.0.5.** `happy-dom` does not fix it on Vitest 3. Lesson text now says: use Vitest >= 4, or `// @vitest-environment node` for those files.
3. `use(promise)` **tests (lesson 04).** `render()` (sync act) around a suspending component never reveals the content (React warns "suspended inside an act scope, but the act call was not awaited"). Verified fix: `await act(async () => { render(...) })`. The lesson example was rewritten.
4. **Unawaited** `expect(...).rejects/resolves` **(lesson 02).** Not "silently passes" in Vitest 3.2.7 (auto-awaited with a warning, failures still fail); fails the test outright in Vitest 5.
5. **Playwright (lessons 09, 12, 15).** `reducedMotion` is not a top-level `use` key (use `contextOptions`); component-testing packages removed in 1.63; Docker tag is `-noble`; Node 18

is not supported (docs: 22/24/26); GitHub Action majors bumped.

6. **TypeScript example in lesson 11 failed to type-check** (`expectTypeOf(ok1.onChange)` is narrowed by assignment); fixed and both forms verified with `tsc --strict` and `vitest --typecheck`.
7. **New majors noted** (short notes, no rewrites): Vitest 4/5, MSW 3, React Router 8, TypeScript 6/7, ESLint 10, jest-dom 7, Playwright 1.64, Storybook 10.

Per lesson

01 Test strategy and pyramid

- Checked: `applyPercentOff(1999, 15)` = 1699 (executed, confirmed). Vitest default file isolation (v3 config docs, confirmed). jsdom lacks layout/ResizeObserver (confirmed by lesson-10 run: no `showModal`, no `inert`).
- **Corrected:** MSW handler `http.post("http://localhost/api/cart/promo")` does not match a jsdom relative fetch (jsdom origin is `http://localhost:3000`); executed, changed to `"*/api/cart/promo"`. Browser-mode "verify maturity" replaced (no "experimental" label in current docs).
- Removed 4 `⚠`.

02 Unit testing with Vitest

- Checked and **confirmed** by running on 3.2.7: `mockReset` returns to the original `vi.fn(impl)` (3.x semantics), `vi.hoisted` + `vi.mock` example (signup), the ReferenceError text when `vi.hoisted` is omitted, `test.each` with `$prop` titles, `test.extend` fixtures (teardown after `use`), debounce fake-timer example (299/1 ms boundaries), `advanceTimersByTimeAsync` vs sync, `toHaveBeenCalledExactlyOnceWith`, `vi.getTimerCount`, `vi.waitFor`, `expect.poll`. Docs (v3, v4, v5 migration guides): default `include`, `testTimeout` 5 s, `restoreMocks` / `mockReset` semantics, `--sequence.shuffle`, typecheck default `*.test-d.ts`, `pool: forks` default, `coverage.all` default true in v3/removed in v4, `workspace` deprecated in 3.2 and removed in 4.
- **Corrected:** un-awaited `.rejects` (see above); default fake-timer list (everything except `nextTick` / `queueMicrotask`, so `performance`, `setImmediate`, `rAF` are faked); `coverage.all` statement; `projects` vs `workspace`; `vitest --typecheck` only checks `*.test-d.ts`; `css: false` comment; RTL cleanup/act setup only registered with global `beforeAll/afterEach`.
- **Added:** version note covering Vitest 4 and 5 breaking changes relevant to the lesson (clearMocks default, hoisted-call errors, `expect.poll` timeout, unawaited assertions fail, `restoreAllMocks` change, constructor mocks, pool options removed).
- Removed 25 `⚠`.

03 React Testing Library

- Confirmed by running: `findBy` default timeout ~1000 ms (measured 1002 ms); `description` option for `getByRole`; `reactStrictMode` option via `render` / `renderHook` / `configure` (RTL 16.3.3; mount = 2 renders, 2 effects, 1 cleanup); email input value is trimmed by jsdom/user-event; React 19 form action + `useActionState` through `userEvent` (input reset after action); `{Enter}` implicit submit; row-by-name query; `toHaveBeenCalledExactlyOnceWith`.

- **Corrected:** `{enter}` lowercase does work in v14 (case-insensitive fallback) while an unknown key like `{foo}` silently does nothing; auto-cleanup/act-environment conditions (RTL `index.js`, confirmed by a `globals:false` run); `listitem` has no name from content; the Playwright-CT table row; React Router import notes (v7/v8); jest-dom 7 note; fake-timer advice (`advanceTimers` alone insufficient).
- Removed 16 `⚠`.

04 Testing async updates

- Confirmed by running: `waitFor` timeout/interval defaults (DTL source); act warning text ("An update to %s inside a test was not wrapped in act(...)") with `IS_REACT_ACT_ENVIRONMENT=true`; no warning inside `findBy`; no warning for updates after unmount; polling example incl. StrictMode; unmount cleanup; delayed fake-timer example; `waitForElementToBeRemoved` error text; error boundary + `use`.
- **Corrected:** fake timers section (see headline 1) including the SearchBox tests (both rewritten with the `jest` stub; mutation check also verified); Suspense/`use` test (headline 3); `--repeat` is not a Vitest CLI flag (use `repeats` test option); act-warning cause table row for "after unmount"; note that RTL only sets the act flag with global `beforeAll`.
- Removed 20 `⚠`.

05 Testing custom hooks

- Confirmed by running: `renderHook` `initialProps` / `wrapper` / `reactStrictMode`, counter batching example, `useDebouncedValue` examples, `vi.getTimerCount`, localStorage spy behaviour (prototype spy works, direct spy records 0 calls), `StorageEvent` with `storageArea`, `expect(() => renderHook(...)).toThrow(...)`.
- **Corrected:** `rerender()` without args re-renders with `undefined` props, it does not reuse previous props (lesson claimed reuse); React 19 does not `console.error` for errors rethrown via act; MSW URL `http://localhost/...` replaced by `*/...` pattern; `useEffectEvent` status (stable in 19.2, exported by 19.3.1).
- Removed 12 `⚠`.

06 Mocking strategies and MSW

- Confirmed by running (MSW 2.15): handler precedence/`once` fall-through, `HttpResponse.error()`, malformed JSON, `delay("infinite")`, stateful todos handlers + `tsc --strict` generics `http.post<never, Body, Resp>`, GraphQL operation matching, `graphql.link`, `ws.link` API, `sse` / `HttpHandler` / `passthrough` / `bypass` exports, `server.events.on/removeAllListeners`, `listHandlers`, `delay()` default ~5 ms in Node, abort in node env (AbortError, handler still completes).
- **Corrected:** relative URLs (work under jsdom while MSW listens; throw `Failed to parse URL` otherwise or in node env); abort section (headline 2, plus happy-dom not a remedy); unhandled-request error text; WebSocket support dates (2.6.0), SSE (2.12.0); `poolOptions` renamed in Vitest 4.
- **Added:** MSW 3.0.0 note (ESM-only, `onUnhandledFrame`, `msw/graphql`, no longer patches `setTimeout`, cookie changes, `websocket:connection`).
- Removed 22 `⚠`.

07 Testing API interactions

- Extracted and executed all of §5.1-5.9 (client.ts, TodoPage, 13 UI tests, 8 client tests). Result: 21/21 on Vitest 4.1.12 after two corrections, 12/21 on Vitest 3.2.7 (headline 2).
- **Corrected:** `e instanceof DOMException` fails under jsdom (Node's `AbortError` is a different `DOMException`); replaced by a name check. The "posts the trimmed text" test failed because the GET fake always returned `[]` and the post-mutation refetch removed the new row; fixed with a stateful `db`. TanStack v5 status/field names and retry defaults confirmed.
- Removed 7 `⚠`.

08 Integration testing

- Ran a router test (guard redirect via `Navigate`, `lazy` route, React 19 `<title>` hoisting to `document.title`, `router.state.location.pathname`, Zustand `getInitialState` reset, persist API methods).
- **Corrected/added:** `createMemoryRouter` fails on Vitest 3.2.7 + jsdom (AbortSignal); with react-router 7.18 under Vitest 4, importing `RouterProvider` from `react-router/dom` while hooks come from `react-router` gave "`<Navigate>` may be used only in the context of a `<Router>`" (two module instances); `RouterProvider` from `react-router` works, and react-router 8.5 (ESM-only) works with `react-router/dom`. Next.js testing guide quote verified. `vitest --changed` confirmed.
- Removed 12 `⚠`.

09 E2E with Playwright

- Type-checked all code snippets (config with `devices`, `webServer`, setup project `dependencies`, `tag`, `clock`, `toPass`, `waitForResponse`, `route`, `AxeBuilder`) with `tsc --strict` against @playwright/test 1.64.1 (only errors: fixtures the snippets reference but do not define). Docs verified: timeouts (30 s test, 5 s expect, no action/navigation timeout), clock API (1.45; methods listed), `storageState` sessionStorage caveat, `tag` option, `grep`, blob + `merge-reports` workflow, `toHaveScreenshot` options, `reducedMotion` location.
- **Corrected:** Node 18+ (docs: 22/24/26); component testing status (removed in 1.63); `reducedMotion` is under `contextOptions`; Vitest `exclude`/Vitest 4 default exclude; action versions.
- Removed 16 `⚠`.

10 Accessibility testing

- Ran vitest-axe 0.1.0 + axe-core 4.14 in jsdom: matcher setup (`vitest-axe/matchers`, `extend-expect`), failing case message, unawaited `axe()` (throws "No violations found in aXe results object"), `color-contrast` lands in `incomplete` in jsdom (so passes), `region` fires on `document.body` scans and `configureAxe` disables it, `HTMLDialogElement.showModal` / `close` / `inert` missing in jsdom 30.
- **Corrected:** the "forgetting await passes vacuously" explanation (it fails with a misleading message); jsdom dialog caveat made concrete; WCAG 2.2 status confirmed (REC 12 Dec 2024). Noted vitest-axe is lightly maintained (last publish Jan 2025).
- Removed 4 `⚠` and several "(verify)" tags.

11 Component contract testing

- Ran Button (ref-as-prop in React 19) and the reusable TextField contract suite (10/10 pass). Ran the type-test example under `tsc --strict` and `vitest --typecheck`.
- **Corrected:** the `expectTypeOf(ok1.onChange)` assertion was wrong (assignment narrowing); fixed. React 19 render errors are not `console.error`-logged. Storybook portable stories API verified in docs (`composeStories`, `Story.run()`). Zod 4 note.
- Removed 9 ⚠.

12 Visual regression

- Verified in docs: `toHaveScreenshot` option names, first-run "A snapshot doesn't exist ... writing actual", platform-suffixed names, `snapshotPathTemplate` tokens (and that the lesson template has no `{platform}`), `--update-snapshots`, `page.clock`, `--repeat-each`.
- **Corrected:** Docker image `v1.XX.X-jammy` -> `v1.64.0-noble`; Vitest browser-mode note.
- Removed 4 ⚠.

13 TypeScript correctness

- Ran the §3.2 tsconfig and React 19 typing claims on TS 5.9.3 and TS 7.0.2 (`useRef<T>()` error, global `JSX` removed, ref-callback implicit return flagged, generic component with `ref` prop, `noUncheckedIndexedAccess`): all confirmed identically. `vitest --typecheck` verified; `vitest typecheck` subcommand is not documented, replaced.
- **Corrected:** `e.target.value` on `ChangeEvent<HTMLInputElement>` compiles (target is `EventTarget & T`); only `FormEvent` / `MouseEvent` need `currentTarget`. **Added:** TS 6/7 note (strict default, removed `baseUrl`, `node10`, etc.), Zod 4.
- Removed 8 ⚠.

14 Linting and static analysis

- Ran the §5.2 flat config with ESLint 10.13: it works, reports hooks/a11y/Testing Library/floating-promise violations. `TIMING=1` verified.
- **Corrected:** `tseslint.config()` is deprecated (use `defineConfig` from `eslint/config`); linting `eslint.config.js` itself with `projectService` fails unless excluded; react-hooks 7 exports `configs.flat.recommended` and compiler-derived rules; jsx-a11y 6.10.2 peer range stops at ESLint 9 (needs `--legacy-peer-deps`); ESLint 10 Node range; `useEffectEvent` stable.
- Removed 7 ⚠.

15 CI checks

- Parsed the 7 YAML blocks (valid YAML; sketches with `...install...` placeholders are intentionally abbreviated). Verified `vitest run --shard`, `--outputFile.junit`, Playwright `merge-reports`, coverage `thresholds` option names.
- **Corrected:** action majors updated to checkout v7, setup-node v7, cache v6, upload-artifact v7, download-artifact v8 (latest releases on 2026-10-22; notes added about Node 24 runtime and runner minimums); Playwright container tag; link URLs checked (200).
- Removed 2 ⚠.

16 Avoiding brittle tests

- Ran the §5.5 SearchBox test: as written it times out under Vitest 3.2.7 (hang); fixed with the `jest` stub; `--sequence.shuffle`, `restoreMocks`, `unstubGlobals` names verified.
- **Corrected:** B3 answer (fake timers), `listitem` name, "Playwright CT" row, TanStack `gcTime`.
- Removed 4 `⚠`.

17 Phase assessment

- Executed B1, B2, B3, B5 (outcomes and `onChange` values `'ab'`, `'ac'` confirmed; StrictMode 2 calls confirmed) and B6 (`TS7030` with `noImplicitReturns`, nothing under `strict` alone).
- **Corrected:** B4 answer (RTL async wrapper + user-event delay; verified fixes), action versions, Docker tag, verify-list remarks. Removed 1 `⚠` plus several "(verify)" tags.

Unresolved (could not verify; text hedged or flagged in-lesson)

1. Pact (`@pact-foundation/pact`) usage in lesson 11 was not run.
2. Stryker (lessons 01, 16): not run; only noted that a Vitest runner exists and Stryker 10 is current.
3. Chromatic/Percy/Storybook test-runner specifics (lesson 12): vendor docs not fetched beyond Storybook portable stories and visual-testing page reachability.
4. No verified recipe for restoring Node's `AbortController` inside Vitest 3's jsdom environment (lesson 06 states this explicitly); only upgrade-to-Vitest-4 and node-environment fixes are verified.
5. React Router 7.18 + Vitest 4 dual-instance cause (lesson 08) is described empirically; the exact resolution-condition cause was not isolated, and Vitest 3 + RR 7 could not be tested separately because of the AbortSignal failure.
6. Vitest 4 behaviour for mock changes (constructor mocks, `restoreAllMocks`) is taken from the migration guide, not executed (only 4.1.12 `AbortSignal`, lesson-07/08 tests and Vitest 5 `unawaited` behaviour were executed).
7. Playwright browsers were not installed, so no Playwright test was actually run; verification was by docs and `tsc` types only.
8. `vitest-axe` TypeScript augmentation (`extend-expect`) was imported at runtime but not type-checked in a full tsconfig.